CCCA Cell-State Analysis — Bassez et al. 2021, Breast

Breast Cancer · 10x · 65,837 malignant cells (41 samples)

GO / NO-GO GATES

Gate 1 — mixing & continuity GO

Harmony(theta=10) integrates all 41 samples: same-sample neighbour fraction 0.37 -> 0.21, iLISI median 5.51 -> 9.10 of 41. One connected component (45 cells / 27 specks removed). Read on the regenerated embedding + per-neighbourhood iLISI, not a scalar: mixed AND continuous.

Gate 2 — module geography GO

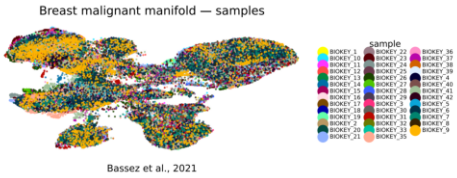
Non-branching case: a single (Gast/pSom)->Crest developmental gradient. Gast & pSom co-vary (smoothed r=-0.21), both oppose Crest (pSom<->Crest r=-0.82, Gast<->Crest r=-0.24). Along pseudotime Gast/pSom fall (rho -0.11/-0.32), Crest rises (rho +0.32) to the terminal lobe. joint pSom&Crest enrichment 0.23 (<1 = inverse gradient, not a co-expression hub).

Gate 3 — cross-repression NA

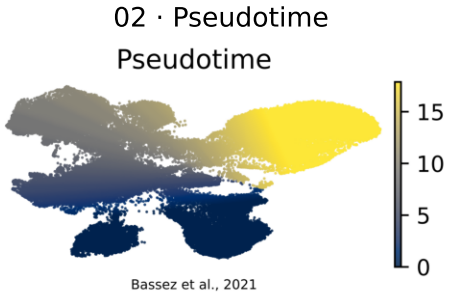
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks). No bifurcation, so the branching-only hub cross-repression test is not run (correlation / modules notebooks are not produced, per the deliverable spec).

STEP FIGURES

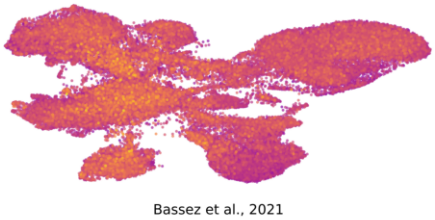
01 · Samples (Gate 1)



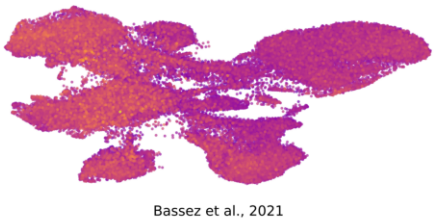
02 · Pseudotime



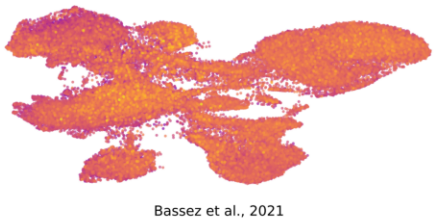
03 · Gastrulation



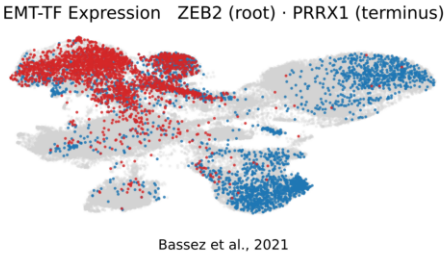
03 · Somitogenesis



03 · Neural Crest



04 · EMT-TFs



Dataset: Breast tumours (Bassez 2021), 10x, 65,837 malignant cells / 41 samples; single low-cycling body (2% cycling).

Verdict: Pattern reproduces in the non-branching form (Linear-pSom/Crest): one continuous gastrulation/somitogenesis -> neural-crest axis, Crest terminal. G1 GO / G2 GO / G3 NA. EMT-TFs ZEB2 (root) -> PRRX1 (terminus).

PARAMETER CHOICES

Sample threshold	geom-mean of 92->3 spillover cliff = 16 (drops 3-cell BIOKEY_34) -> 41/42 samples
Overdispersed genes	2,498 (scFates find_overdispersed)
Integration	Harmony on sample, theta=10 (mixing flat across sweep; strongest for patient batch)
Embedding	n_pcs=20, n_neighbors=30, spread=1.3; 45 cells / 27 islands removed -> 65,837
Geometry	Non-branching -> scFates principal curve (ElPiGraph), Nodes=20, lambda=0.3, mu=100
Topology check	2 tips / 0 forks; fold self-approach 2.53 (>1 fold-free); pt vs prolif rho -0.24
Root	tip 2 = Gast/pSom-high, Crest-low developmental origin (from signature order)
Module means (raw)	Gast -0.014 · pSom +0.022 · Crest -0.005 (Linear-pSom/Crest)
EMT-TFs shown	ZEB2 (root, rho -0.07) <-> PRRX1 (terminus, rho +0.07); detection-gated, FOXC/PRRX in panel